
BEVY_ZEROVERSE: AUDITABLE PROCEDURAL INDOOR DATA FOR MULTI-VIEW RECONSTRUCTION

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ABSTRACT

We describe an open-source synthetic data engine combining primitive-object scenes with furnished procedural interiors, calibrated multi-view cameras, optional human motion, and aligned geometric annotations. Indoor generation samples continuous architectural, furnishing, appearance, lighting and camera parameters under geometric constraints. A continuous camera-baseline control coordinates spacing, shared-surface overlap and path variation. The current generator-21 evaluation covers 2,048 consecutive layouts and 512 distinct rendered rooms. A matched room subset measures co-visibility distributions at five baseline values. The report presents absolute measurements of the current generator, with exact configurations and reproducible artifacts. Photographic realism, unlimited-process stability and downstream pretraining gains are not established by these diagnostics.

1 Purpose and scope

Sparse-view reconstruction benefits from observations of shared surfaces: cameras that see unrelated objects offer little direct correspondence supervision. LRM-Zero demonstrates the potential of procedurally synthesized object data [1]; MVSplat uses cross-view matching to estimate geometry [2]. Our contribution is an inspectable data-generation system and its bounded evaluation, rather than a trained reconstruction model or a reconstruction accuracy result.

Version 0.25 retains the existing Object, CornellCube, Room, SemanticRoom and Human scene modes, and exposes furnished interiors separately as `ProceduralIndoor`. The runtime uses Bevy 0.19.1, optional Burn 0.21 inference, and published AnnyBody/ARDY integrations. Office, conference, lounge, classroom, library and related activity priors influence a compositional program; they do not select complete meshes or a finite collection of room templates. Most architecture, furniture, plants and clutter require neither mesh repositories nor photographic texture collections. AnnyBody human meshes and optional motion/embedding models are explicit pretrained dependencies.

2 Procedural scene program

A scene is generated as $S = G(s, \theta)$, where s seeds independent random streams and θ contains user-controlled densities, activity, quality, camera and motion policies. The serialized manifest records the sampled program, object transforms and seeds, people, photometry and camera paths. A separate generator version and capture-engine identity invalidate incompatible resummptions. Determinism means reproducing the same program and settings with compatible assets and engine versions; bit-identical rasterization across GPU vendors is not promised.

2.1 Architecture, objects and appearance

Room area and aspect are sampled logarithmically; width and depth are clamped to 5.6–21 m and height spans 2.65–4.8 m. Furnishing zones have continuous bounds, orientation, aisle width, occupancy and mixed activity weights. Partitions sample portal width, thickness, glazing fraction, mullion spacing and optional transoms. Pillars, ceiling relief, paneling, reveals, trim and shelves add geometric structure. One to three exterior faces independently receive glazing; the fourth

face retains the neighboring-room connection. Window opening dimensions, sill/head heights, bay widths, frames, inset depth and shades vary continuously. Rough openings are subtracted from the solid wall geometry; decorative surfaces respect these openings. The exterior envelope remains rectangular: these are varied rectilinear interiors, not a general architectural CAD system.

Furniture is assembled from material-specific parts. Tables vary outline, taper, thickness and leg/pedestal construction; chairs vary seat shell, curvature, back proportions, recline, arms, base and headrest. Sectional seating, shelving, potted plants, outlets, switches, ceiling lights, whiteboards, displays and tabletop objects extend the semantic inventory. Placement uses support relationships, clearance checks, zone-dependent furnishing fields and bounded rejection. Object yaw and arrangement are sampled rather than restricted to orthogonal grids. Collision proxies approximate detailed meshes; geometric validity is checked separately from physical plausibility.

Surfaces use 35 role-specific base material slots with procedural albedo, normal and roughness maps, metric UV scales and mip chains. Recipes vary spatial frequency, directional grain, textile structure, mineral veining, relief, wear, color and roughness. Object finish variants preserve their underlying maps while varying appearance; screens, boards and printed pages have independently seeded content. Only referenced finish variants are built during runtime preparation. Glass recipes vary roughness, refractive index, tint, attenuation and thickness. Direct lights vary placement, intensity, color temperature, circuit activity and direction; exposure, daylight and sky radiance are independently recorded. Low-light samples are retained in evaluation.

People retain AnnyBody geometry with continuous phenotype/pose parameters, fitted garments, surface hair and facial details. Static people remain the default. Opt-in ARDY motion combines seeded procedural text sequences with object-aware waypoint planning and scene-time playback. Requesting static scenes does not initialize motion inference. Motion models use the upstream loaders and caches; cached prompt text does not replace the motion seed. Clothing, hair, joint limits and collision handling are approximations, particularly during complex motion.

2.2 Multi-view sampling and training defaults

Camera centers and complete trajectories are constrained to the primary furnished room by default. The sampler varies focal length, position, target, roll and path shape, checks obstacle clearance and prefers visible content. Static capture uses one timestep; optional trajectories support repeated calibrated observations. A reference camera anchors a connected set of views. For each other camera and five trajectory times, we require

$$\min\{o_{0 \rightarrow i}(t), o_{i \rightarrow 0}(t)\} \geq \tau, \quad b_{\min} \leq \|c_i(t) - c_0(t)\| \leq b_{\max},$$

where o estimates first-surface overlap using visibility rays and object/architecture proxies. Overlap constrains reference edges. An additional minimum separation applies to every camera pair at 33 synchronized path samples. Group spread is the horizontal camera-center covariance’s minor/major standard-deviation ratio. Moving pairs must differ in normalized RMS displacement after removing their starting offsets. Full-path swept collision checks remain; sampled spread and overlap are not continuous-time or rendered-pixel guarantees.

The camera-baseline control $b \in [0, 1]$ expands into explicit advanced settings: minimum pair separation $0.04 + 0.46b$ metres, minimum reference distance $0.06 + 2.30b$ metres, maximum reference distance $0.35 + 9.65b$ metres, overlap target $0.65 - 0.55b^{0.6}$, group spread $0.20 + 0.10b$, and independent path variation b . The default is $b = 0.5$. Wide settings permit distant viewpoints and lower overlap; narrow settings favor close views of shared content. Collision and visibility rejection affect the realized distribution. Arbitrary room, camera-count and constraint combinations need not be feasible.

The inspector exposes this slider above the advanced controls. Any advanced edit switches to explicit custom values. Changes take effect on regeneration, so dragging a control does not rebuild the room. JSON input accepts `{"baseline":0.75}` within the `indoor_camera` configuration. Serialization records the expanded values, avoiding hidden preset precedence. Camera travel length and long-route frequency are separate from spacing between cameras. Joint collision and overlap rejection can favor shorter accepted routes at wide spacing; the minimum travel bound remains explicit. The finite search budget reports impossible settings without silently weakening them.

3 Capture and annotation contract

Capture begins only after scene preparation, assets, shader pipelines, lighting and requested human trajectories are ready. A bounded asynchronous readback protocol associates outputs with the requested scene, camera, time and mode; it does not infer readiness from an arbitrary number of frames. Geometric channels share an RGBA32Float ground-truth raster attachment. RGB uses Bevy’s PBR pipeline, shadow maps, tone mapping and optional indirect-lighting approximation.

Indoor training configuration	Default and rationale
Capture cameras / timesteps	Four / one; synchronized views of static geometry
Shared-view policy	$b = 0.5$; $\tau \approx 0.287$, pair separation ≥ 0.27 m, reference distance 1.21–5.175 m; explicit <code>multiview:null</code> enables independent views
Group diversity	Horizontal spread ratio ≥ 0.25 for 3+ views; moving-pair normalized RMS difference ≥ 0.075
Room/activity/density	Primary room, Mixed activity, furnishing density 0.65, human density 0.25
Camera path policy	0.03–8 m allowed path length, 0.45 long-path proposal fraction; collision rejection still applies
Human motion	Disabled; static people remain, no ARDY initialization
Native lighting	Auto quality, shadows and approximate baked diffuse GI, 256 rays/probe
Dataset CLI workers / chunks	One process, 16 rooms/chunk; explicit overrides retained
Finite worker lifetime	256 scenes/child for finite indoor jobs with process workers
Default export	RGB and calibration; geometric channels are opt-in

Table 1: Defaults conditional on the procedural-indoor dataset CLI scene. Other scene modes retain their defaults. These are an engineering starting point, not an optimum established by downstream training. Library callers explicitly choose camera and worker counts.

Output	Meaning and qualification
Color	Display RGB; lossless float storage or explicitly chosen codecs. Separate linear-HDR export supports renderer comparisons.
Depth	Colorized, normalized or metric linear depth; axial/ray conventions; calibration records the selection.
Normal / position	View-space normals encoded as $(n + 1)/2$; world positions affine-normalized by the exported AABB, without clamping. First surface, float32 geometric path.
Semantic	Shared class vocabulary, aligned with depth/normal/position, including ceiling light fixtures.
Optical flow	Forward displacement $(\Delta u, \Delta v)$ in pixels, with correspondence-valid and target-visible masks. Terminal time has no forward correspondence.
Motion vectors	The same exported displacement divided by image width/height. Viewer velocity colors are previews, not these numeric annotations.
Co-visibility	Optional same-time, per-pixel membership in other capture cameras. A uint16 mask assigns one bit per camera (up to 16), with a separate source-valid mask and a consistent additive RGB legend; the source camera and editor are excluded.
Calibration / instances	Camera intrinsics, poses and normalized trajectory time, scene manifest, semantic instance oriented boxes, person joints, and primary-room annotation AABB.
O-Voxel	Primary-room architecture/objects only; excludes neighboring/outside geometry. Requires one timestep and disabled human motion; incompatible requests are rejected.

Table 2: Eight currently supported render modes plus structured annotations. The population experiment exports five modes and derives co-visibility independently from depth and calibration; the gallery exports all eight modes. Glass is a first geometric surface: annotations do not follow refracted or reflected light paths. Flow transports surface correspondence, including supported human deformation, rather than estimating photometric image motion.

Float32 storage does not imply that all RGB or lighting computations have float32 intermediate precision: HDR color and irradiance volumes include float16 stages.

Headless generation supports chunked safetensors or per-sample files, compression, resumable identity checks, metrics and bounded worker replacement. RGB and numeric channels have separate encoding contracts. Native generation performs numeric temporal export and O-Voxel export. The WebGPU viewer supports indoor construction, inspection and optional motion inference, using the same manifest and geometry modules; Portable quality reduces unsupported or costly rendering features. Browser velocity visualization is not a qualification of native dataset flow export. The current Wasm path omits native baked diffuse GI and uses its platform lighting fallback.

4 Current-generation evaluation

The current generator is audited on 2,048 consecutive layout seeds (24000–26047) at furnishing density 0.65, static human density 0.25, four cameras and default baseline $b = 0.5$. All layouts pass geometry, swept collision and sampled

camera constraints. The population contains 508 numeric series, 406 with nonzero observed variance. These are correlated measurements.

The rendered population contains 512 distinct consecutive rooms (24000–24511), four 320×240 cameras and times 0, 0.5, 1. The first 128 rooms are additionally captured at $b = 0, 0.25, 0.75, 1$, yielding 1,024 room/configuration captures and 12,288 views. Every row in the parameter sweep uses the same 128 rooms, with identical geometry, people, materials and lighting, verified by manifest hashes. No failed, dark, or low-overlap sample is replaced. Native Auto lighting and shadows are enabled; baked diffuse GI is disabled for the sweep. The selected gallery uses a separate full-lighting recipe. Captures use the checkout-local wgpu command-cache/upload optimizations; published crates resolve registry wgpu. Timing observations do not qualify unpatched package performance.

Baseline	Distance (m)	Overlap (%)	Shared (%)	All three (%)	Proxy misses
0.00	0.19	72.8	92.1	61.0	177 / 1152
0.25	1.33	55.2	85.2	39.9	96 / 1152
0.50	2.19	46.5	80.7	28.4	52 / 1152
0.75	2.94	38.9	76.9	22.0	22 / 1152
1.00	3.60	32.9	72.8	17.3	11 / 1152

Table 3: Absolute measurements at five settings of the current camera program. Distance is the median room-mean reference distance, measured at 33 times. Overlap averages bidirectional reference pairs at three rendered times. Shared and all-three fractions use valid source-pixel observations. Each row contains 128 rooms, 1,536 views and 1,152 reference pairs; its proxy threshold depends on baseline.

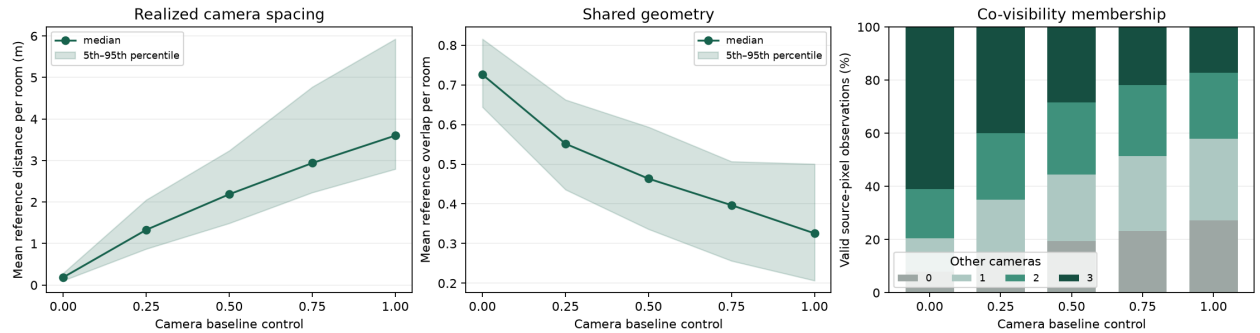


Figure 1: Realized spacing, room-level overlap quantiles and co-visibility cardinality versus the camera-baseline parameter. Pixel observations and views from the same room are correlated.

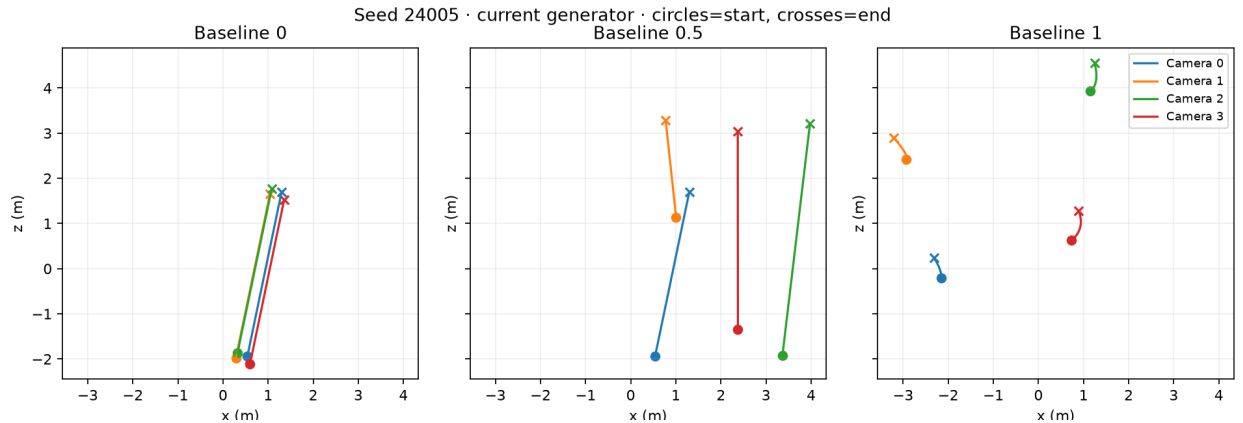


Figure 2: Current camera programs for one fixed room at three baseline settings, using common metric axes.

At default baseline, all 512 rooms contribute 6,144 views and 4,608 reference-pair/time observations. Mean reference overlap is 45.7%, with minimum 16.0%; 210 pairs miss the proxy target. 80.3% of valid source-pixel observations are visible to another camera, and 27.9% to all three others. The denominator is 471,859,200 valid pixel observations, not unique 3D points. Membership excludes the source camera.

The independent diagnostic requires nearest-target-pixel depth agreement within $0.01 \text{ m} + 0.002z$. The production GPU annotation uses a separate symmetric tangent-plane predicate; its exact masks are exported with the gallery. Both use the first geometric surface, treating glass as annotation-opaque. Reflections and refractions are not counted as geometric correspondence.

There are 0 rooms with a view containing at most two semantic classes, and 0 of 357 occupied primary rooms have no person pixels in any captured view. Maximum per-view p99 depth/position disagreement is 0.0000038 m . These are annotation and coverage checks, not photographic-realism metrics. Complete distributions, source identities, configurations and commands are in `docs/camera_baseline_v21.md`. Bounded capture processes do not establish unlimited-process memory stability.

4.1 Seeing the baseline control in a fixed room

Figures 3 and 4 show the visual effect of camera-baseline sampling in two fixed scenes. Each column contains all four synchronized RGB views and a top-down camera plan at $t = 0$. Columns change only the camera policy: geometry, furnishings, static people, materials and lighting are identical within a room, verified by hashing all noncamera manifest fields. The project page provides five captured settings ($b = 0, 0.25, 0.5, 0.75, 1$), an interactive four-view comparison, and a shared-surface overlay backed by exact exported GPU masks. The figures select three of these settings for legibility; they do not select or crop individual camera views.

These selected illustrations use native Auto lighting, 1024 diffuse-GI rays per probe and 768×480 images. They are separate from the consecutive population experiment above. The ten room/configuration captures yield 40 RGB views at one time; the two default groups reuse the matched gallery’s $t = 0$ exports. Image conversion uses a fixed sRGB transfer after renderer tone mapping, with no per-image exposure normalization.

Camera indices identify members of a resampled group, not fixed physical viewpoints across columns. The reference pose and sampled intrinsics may change as rejection finds a feasible group. Hence the examples illustrate the complete baseline policy, not translation of a fixed calibrated rig. Plans retain identical metric axes within each room and show furniture footprints as proxies; short wedges indicate viewing direction, not full frusta or occlusion. Column headers give the mean camera-0 distance at this timestep and the fraction of valid source-pixel observations shared with at least one peer, pooled across all four views. The latter uses the production GPU surface test, distinct from the population depth diagnostic. Pixel observations repeat surfaces; glass remains a first geometric surface.

4.2 Diversity diagnostics

The population export retains zero-preserving object counts, parameter distributions, camera intrinsics and extrinsics, placement heatmaps, correlations and occupancy signatures. These are correlated projections of the procedural program, not independent degrees of freedom. For categorical frequencies p , the entropy-effective count is $\exp(-\sum_j p_j \log p_j)$. The occupancy signature uses a 12-by-12 object-category grid and quantized partition geometry, excluding random seeds and textures. Its HyperLogLog estimate is a bounded geometric redundancy diagnostic, not proof of ten million useful examples.

A seed’s bit width does not establish dataset diversity. Continuous parameter ranges have no useful finite cardinality before quantization; topology, placement constraints, correlations and rendering reduce distinguishable outcomes. Optional frozen SigLIP2 embeddings [3] support image-space redundancy audits, with scene-level aggregation and cross-room nearest neighbors. No embedding-rank or downstream-learning claim is made for this camera-baseline sweep.

5 Matched gallery and viewer

The accompanying project page presents selected generator-21 renders separately from the population evaluations. Two example rooms (seeds 24005 and 24000) were captured at 768×480 , with four synchronized cameras, three normalized trajectory times and all eight render modes. All mode switches reuse these exact scene/camera/time identities. The native CLI used Auto quality, static human density 0.25 and 1024 indirect-lighting rays per probe. Depth/position reconstruction and camera reprojection checks validate alignment before generating website previews. Display mappings for depth, normals, positions and temporal vectors are declared alongside the visualizations; they do not replace the lossless numeric exports.

Co-visibility shares the float32 surface raster used by the other geometric annotations. A GPU pass projects each source surface into the other cameras and checks frustum bounds and symmetric local tangent-plane agreement. The tolerance is $\max(1 \text{ mm}, 10^{-4}d)$ at target depth d . The exact membership mask is separate from its additive color preview and

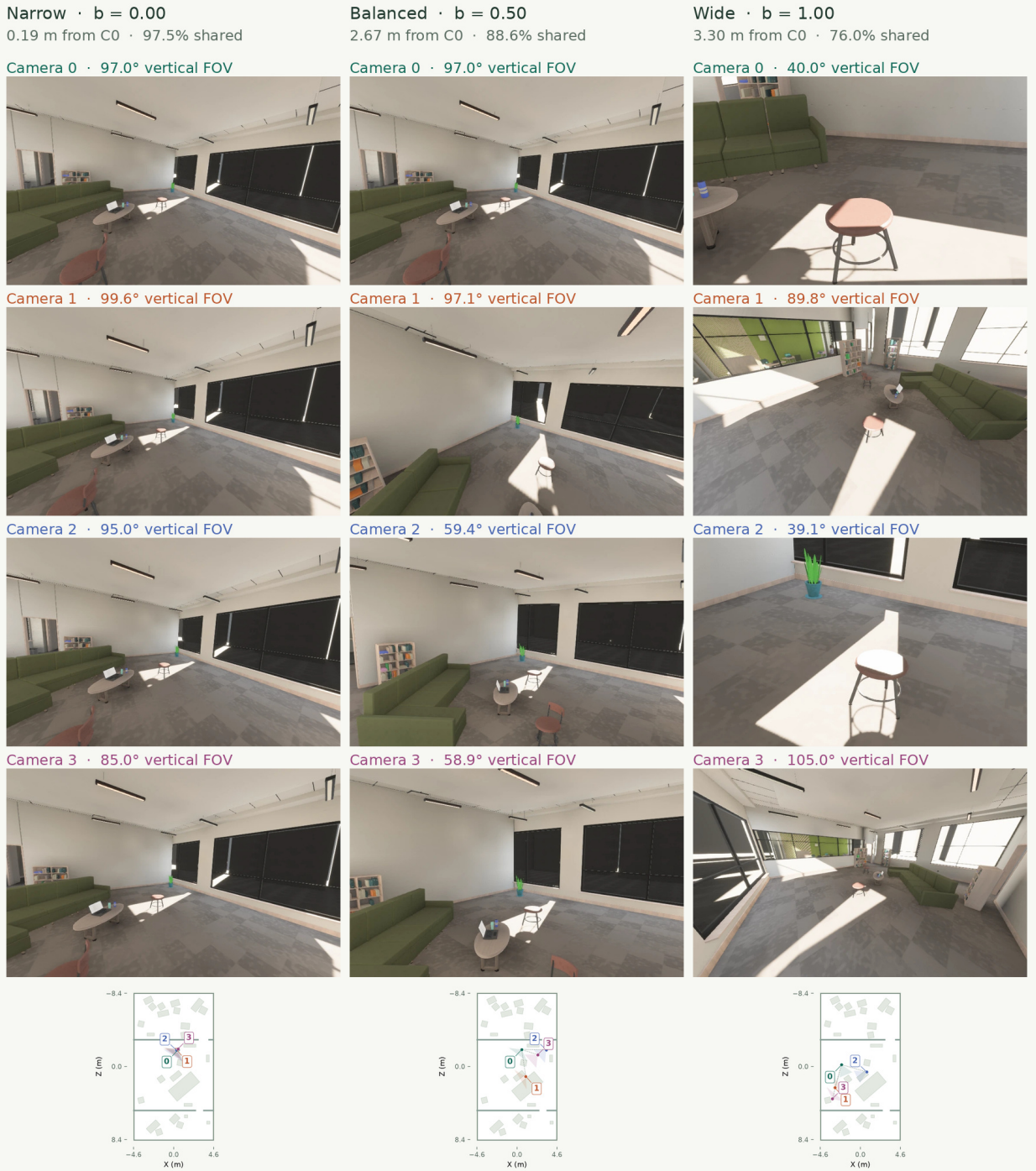


Figure 3: Fixed sunlit seating scene (seed 24005), with narrow, balanced and wide baseline policies from left to right. Rows retain cameras 0–3 in order; the bottom row locates their centers in the same primary room. Close views concentrate on similar content; wider sampling changes perspective and the surfaces jointly observed. Header values describe these particular captures, not population averages.



Figure 4: Fixed workspace scene (seed 24000), using the same column and camera ordering as Figure 3. The room and its lighting are held fixed across baseline values. Feasibility depends on furniture, partitions, camera intrinsics and overlap constraints, so the control is not a requested distance in metres and individual realizations need not vary monotonically.

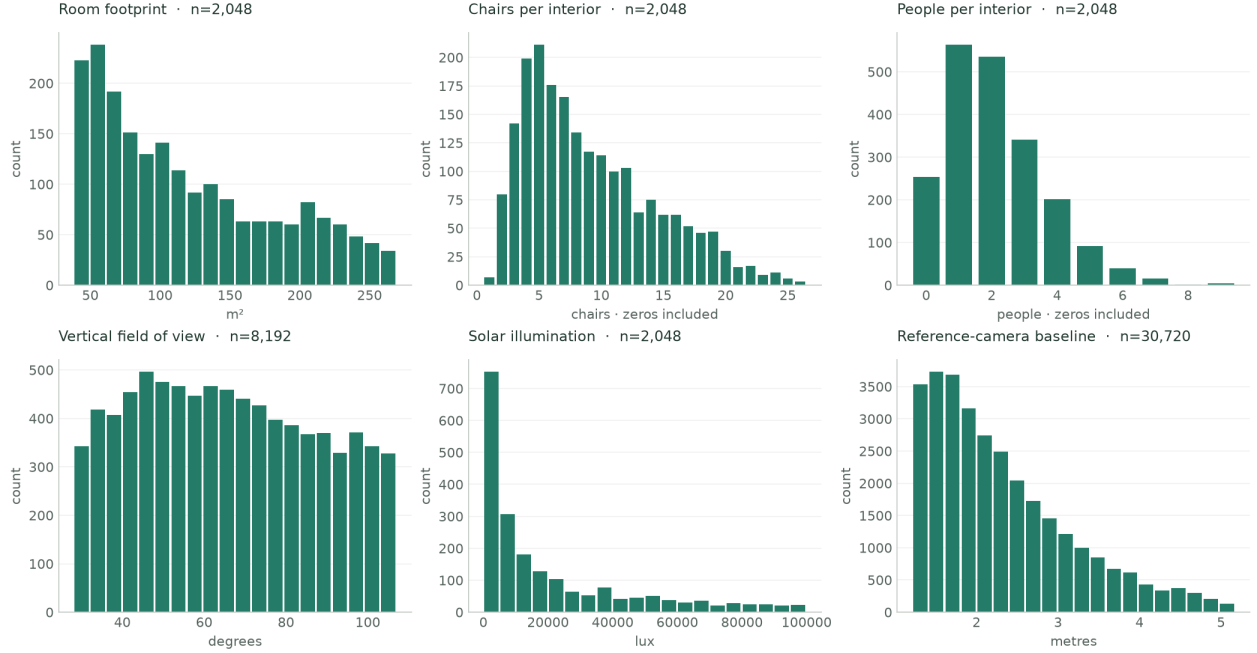


Figure 5: Absolute distributions in the current 2,048-room population at the default camera baseline. Each panel preserves its denominator and units.

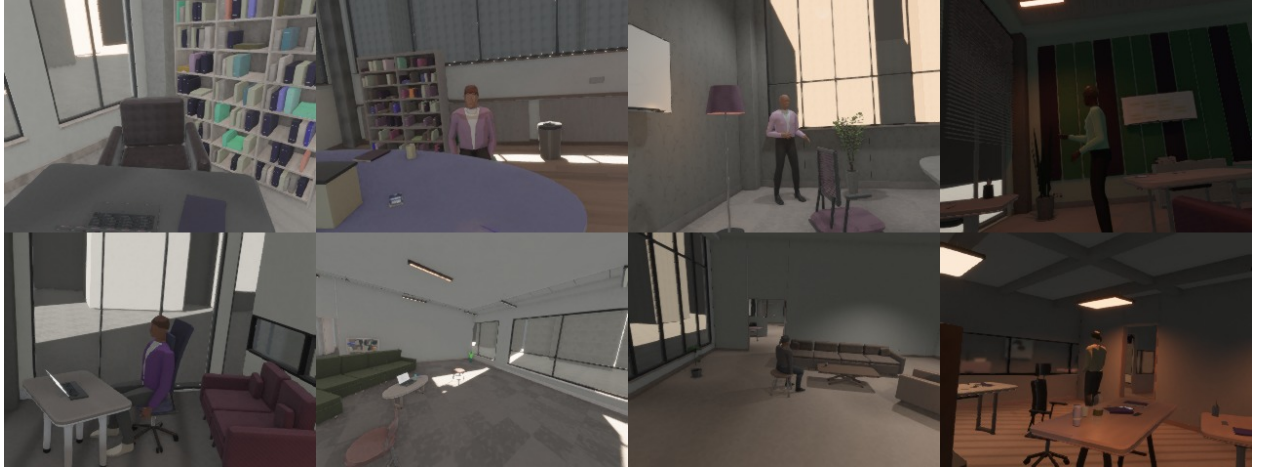


Figure 6: Camera-zero RGB for the first eight consecutive rendered rooms at the default baseline, without aesthetic filtering.

source-valid mask; a valid but unshared surface differs from background even though both have a zero membership mask. This measures raster surface visibility, with the same annotation-opaque glass policy, rather than reflected or refracted RGB content.

Human clothing uses body-derived cross-sections, localized folds and fitted garment details. Viewer pose joints can show through people while environment depth occludes them; editor projections persist through regeneration. ARDY controls expose diffusion steps, history length, guidance and path conditioning. The project page’s separately identified motion example contains two accepted trajectories, two cameras and 120 frames at the model’s 20 Hz rate. It is an illustration of the optional motion pipeline, not part of the static camera-baseline experiment or a motion-quality benchmark.

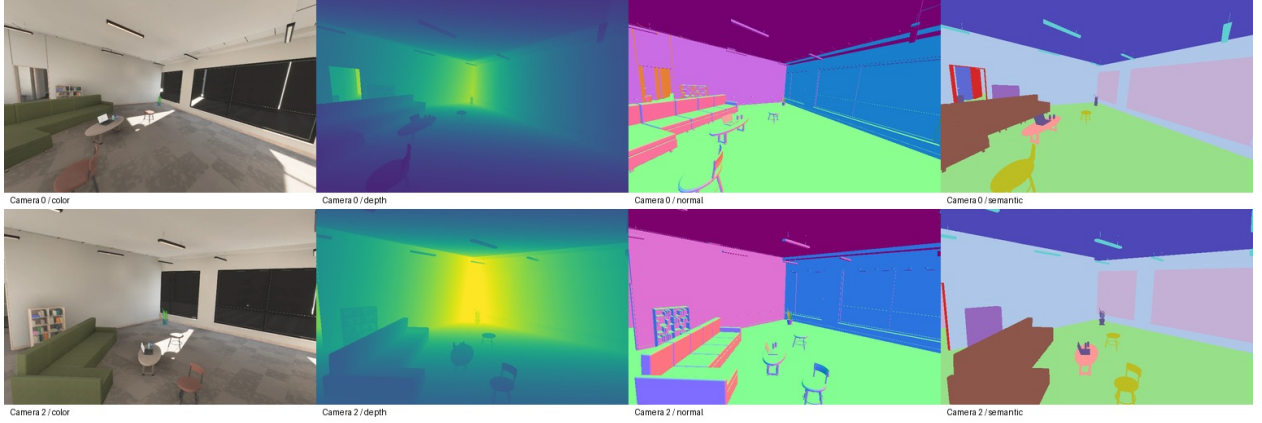


Figure 7: Selected generator-21 room, seed 24005: RGB, metric-depth visualization, view-space normals and semantic labels for cameras 0 and 2 at the same time. Camera selection and pixel-aligned annotation comparisons are interactive on the project page. These illustrations are separate from the consecutive population evaluation.

6 Limitations and remaining evidence

Procedural correctness, broad parameter variation and efficient capture do not establish photographic realism. Hair is a surface groom, skin lacks a full subsurface model, some furnishings remain sparse, object motion uses approximate collision proxies, and diffuse GI uses probes and simplified transport. Glass and lighting are approximations rather than full path tracing. Physical transport accuracy and visual plausibility require independent path-traced reference studies; the camera-baseline sweep does not measure either.

The current study is a generator audit and a small capture cohort, not a ten-million-scene run. It does not establish long-run resident-memory bounds, balanced coverage of all rare architectural combinations, collision-free arbitrary motion, suitability for every reconstruction loss, or transfer to real photographs. Shared-view defaults deliberately correlate cameras within a room; train/validation splits should therefore be made by room seed, not by image. A downstream assessment should freeze a training budget and real evaluation set, compare matched data quantities and camera policies, and measure reconstruction/calibration quality alongside redundancy. The manifests and metrics make that experiment reproducible without claiming its outcome in advance.

References

- [1] D. Xie et al. LRM-Zero: Training Large Reconstruction Models with Synthesized Data. NeurIPS, 2024. <https://arxiv.org/abs/2406.09371>.
- [2] Y. Chen et al. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. ECCV, 2024. <https://arxiv.org/abs/2403.14627>.
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